

# END TERM EXAMINATION

## THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: ES-201

Subject: Computational Methods

Time: 3 Hours

Maximum Marks: 75

**Note:** Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit. Assume missing data if any. Scientific calculator is allowed.

| Q. No.       | Question (Attempt all questions from Q1)                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks      |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|-------|-------|-------|-------|------|-----|--------------|-------|-------|-------|-------|-------|-------|-------|-----|
| 1(a)         | The height of an observation tower was estimated to be <b>47m</b> , whereas its actual height was <b>45m</b> . Calculate the percentage relative error in the measurement.                                                                                                                                                                                                                                                                                                             | [1.5]      |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 1(b)         | Derive the formula for evaluating <b>1/N</b> by Newton's Raphson method.                                                                                                                                                                                                                                                                                                                                                                                                               | [1.5]      |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 1(c)         | Differentiate between partial and complete pivoting in solving linear system of algebraic equations.                                                                                                                                                                                                                                                                                                                                                                                   | [1.5]      |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 1(d)         | Estimate the missing term in the following table:<br><table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>f(x)</td><td>1</td><td>3</td><td>9</td><td>—</td><td>81</td></tr></table>                                                                                                                                                                                                                                                                     | x          | 0     | 1     | 2     | 3     | 4     | f(x) | 1   | 3            | 9     | —     | 81    | [1.5] |       |       |       |     |
| x            | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1          | 2     | 3     | 4     |       |       |      |     |              |       |       |       |       |       |       |       |     |
| f(x)         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3          | 9     | —     | 81    |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 1(e)         | Define cubic spline function and state the conditions required for cubic spline interpolation.                                                                                                                                                                                                                                                                                                                                                                                         | [1.5]      |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 1(f)         | Find $\int_0^{\pi/2} e^{\sin x} dx$ using Simpson's $\blacksquare$ rule for $n = 6$ .                                                                                                                                                                                                                                                                                                                                                                                                  | [2.5]      |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 1(g)         | Solve the IVP $dy/dx = x e^y$ , $y(0) = 0$ using Picard's method and estimate $y(0.2)$ .                                                                                                                                                                                                                                                                                                                                                                                               | [3]        |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 1(h)         | Determine which of the following equations are elliptic, parabolic and hyperbolic:<br>(i) $f_{xx} + 6 f_{xy} + 9 f_{yy} = 0$ (ii) $f_{xx} - 2 f_{xy} + 2 f_{yy} = 2x + 5y$                                                                                                                                                                                                                                                                                                             | [2]        |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| UNIT-I       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 2(a)         | Find the root of the equation $x = e^{-x}$ correct to three decimal places by Secant method by performing six iterations.                                                                                                                                                                                                                                                                                                                                                              | [8]        |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 2(b)         | Use Newton-Raphson method to obtain a root of $\sin x = 1 - x$ to three decimal places.                                                                                                                                                                                                                                                                                                                                                                                                | [7]        |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 3(a)         | [OR] Perform four iterations of Golden section search method to minimize $f(x) = x^4 - 14x^3 + 60x^2 - 10x$ , $x \in (0, 2)$ with $\epsilon = 10^{-3}$ .                                                                                                                                                                                                                                                                                                                               | [7.5]      |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 3(b)         | Use steepest descent method for 3 iterations on $f(x_1, x_2, x_3) = (x_1 - 4)^4 + (x_2 - 3)^2 + 4(x_3 + 5)^4$ with initial point $X^{(0)} = [4, 2, -1]^T$ .                                                                                                                                                                                                                                                                                                                            | [7.5]      |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| UNIT-II      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 4(a)         | Prove that $\Delta = \frac{1}{2} \delta^2 + \delta \sqrt{1 + \frac{1}{4} \delta^2}$ .                                                                                                                                                                                                                                                                                                                                                                                                  | [7]        |       |       |       |       |       |      |     |              |       |       |       |       |       |       |       |     |
| 4(b)         | The table gives the distance in nautical miles of the visible horizon for the given heights in feet above the earth's surface:<br><table><tr><td>x (height)</td><td>100</td><td>150</td><td>200</td><td>250</td><td>300</td><td>350</td><td>400</td></tr><tr><td>y (distance)</td><td>10.63</td><td>13.03</td><td>15.04</td><td>16.81</td><td>18.42</td><td>19.90</td><td>21.27</td></tr></table><br>Find the value of y when $x = 410$ using Newton's backward interpolation formula. | x (height) | 100   | 150   | 200   | 250   | 300   | 350  | 400 | y (distance) | 10.63 | 13.03 | 15.04 | 16.81 | 18.42 | 19.90 | 21.27 | [8] |
| x (height)   | 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 150        | 200   | 250   | 300   | 350   | 400   |      |     |              |       |       |       |       |       |       |       |     |
| y (distance) | 10.63                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 13.03      | 15.04 | 16.81 | 18.42 | 19.90 | 21.27 |      |     |              |       |       |       |       |       |       |       |     |

[P.T.O.]

| Q. No.   | Question (Continued)                                                                                                                                                                                                                                      | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5(a)     | [OR] Evaluate the integral $\int_0^6 [x / (1 + x)] dx$ using Simpson's ■ rule and Simpson's ■ rule respectively. Compare it with the exact value.                                                                                                         | [7]   |
| 5(b)     | Evaluate the integral $\int_0^{\pi/2} [x / \sin x] dx$ using Romberg's method, correct to three decimal places.                                                                                                                                           | [8]   |
| UNIT-III |                                                                                                                                                                                                                                                           |       |
| 6(a)     | Investigate the values of $\lambda$ and $\mu$ so that the system of equations:<br>$2x + 3y + 5z = 9$ , $7x + 3y - 2z = 8$ , $2x + 3y + \lambda z = \mu$<br>have (i) no solution, (ii) unique solution, (iii) an infinite number of solutions.             | [7]   |
| 6(b)     | Solve the system of equations:<br>$x_1 + 10x_2 - x_3 = 3$ , $2x_1 + 3x_2 + 20x_3 = 7$ , $10x_1 - x_2 + 2x_3 = 4$<br>using Gauss elimination with partial pivoting.                                                                                        | [8]   |
| 7(a)     | [OR] Solve the system of equations using Doolittle factorization method:<br>$3x + 2y + z = 4$ , $2x + 3y + z = 5$ , $3x + 4y + z = 7$                                                                                                                     | [9]   |
| 7(b)     | Determine the numerically dominant eigenvalue and eigenvector of the matrix:<br>$A = \begin{bmatrix} 15 & -4 & -3 \\ -10 & 12 & -6 \\ -20 & 4 & -2 \end{bmatrix}$ using Power method. Take initial vector $X^{(0)} = [1, 1, 1]^T$ .                       | [6]   |
| UNIT-IV  |                                                                                                                                                                                                                                                           |       |
| 8(a)     | Employ Taylor's method to obtain the approximate value of $y$ at $x = 0.2$ for the differential equation $dy/dx = 2y + 3e^x$ , $y(0) = 0$ . Compare the numerical solution obtained with the exact solution.                                              | [7]   |
| 8(b)     | Using Runge-Kutta method of fourth order, solve $dy/dx = (y^2 - x^2) / (y^2 + x^2)$ with $y(0) = 1$ at $x = 0.2, 0.4$ .                                                                                                                                   | [8]   |
| 9(a)     | [OR] Solve the initial value problem $y' = x^2 + y^2$ , $y(0) = 1$ , using Euler's method in $0 \leq x \leq 0.8$ , with $h = 0.2$ . Compare the results with exact solution at $x = 0.8$ .                                                                | [8]   |
| 9(b)     | Solve the partial differential equation $u_{xx} + u_{yy} = x + y + 1$ , $0 \leq x \leq 1$ , $0 \leq y \leq 1$ , with $u = 0$ on the boundary numerically up to three iterations with $h = \blacksquare$ . Obtain results correct to three decimal places. | [7]   |

\*\*\*\* End of Question Paper \*\*\*\*